

A serene sunset scene over a pond. The sky transitions from a deep blue to a warm orange glow near the horizon. Silhouetted trees line the background and the right side of the frame. In the center-left of the pond, a fountain sprays water upwards, illuminated from below by a bright yellow light that reflects on the water's surface. The overall mood is calm and peaceful.

Stress

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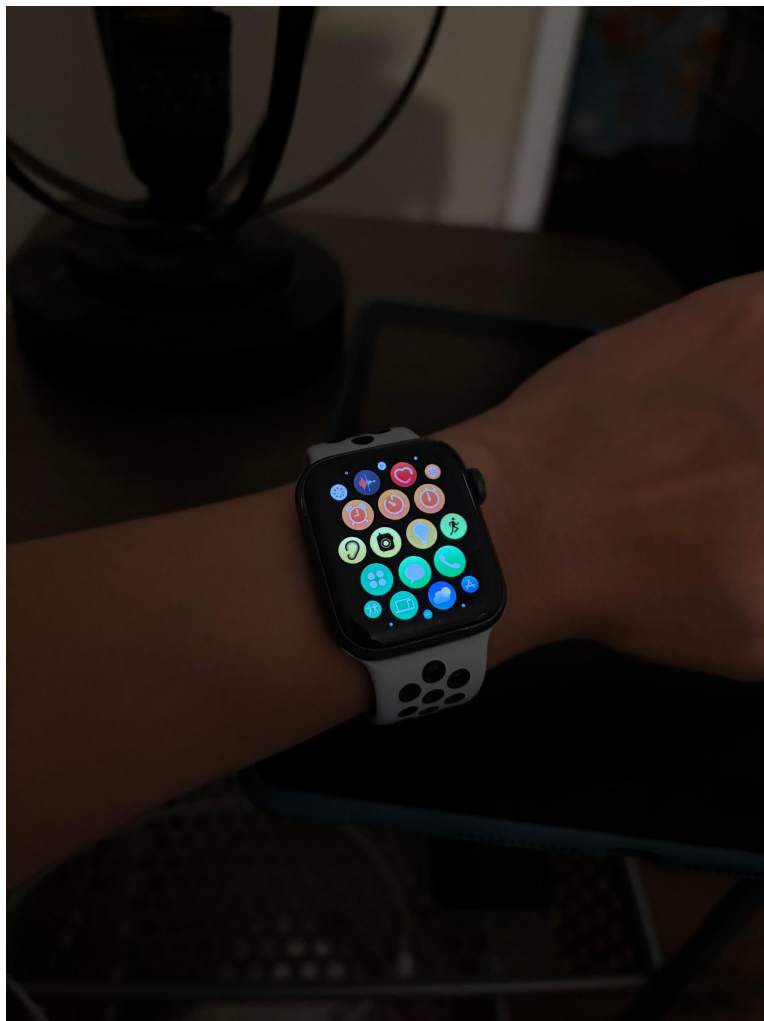
Inspo

- GA tech news article
- Garmin

Problem?



Solution.



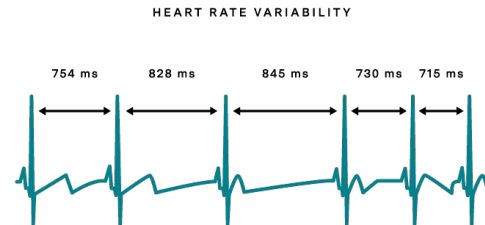
Stress Metrics

- HRV = hr speed + time (Garmin, Apple, Fitbit)
 - Variance in time *between* heart beats (aka RR interval on ECG)
 - RR interval = stress and recovery
 - High HRV, low stress = parasympathetic autonomic nervous system
 - -|-|-|
 - Low HRV, high stress = dominant sympathetic autonomic nervous system
 - -|-|-|

- Sleep (Garmin, Apple, Fitbit)
 - Hours slept
 - Stages of sleep (ex: REM, deep sleep, light sleep, awake)

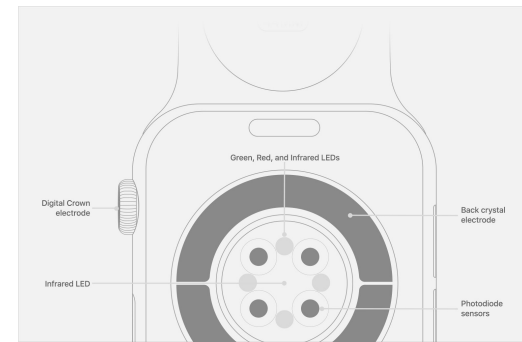
- Body battery (Garmin)
 - Body exertion
 - Composite of HRV, stress, sleep, physical activities

- Sources
 - <https://www.garmin.com/en-US/garmin-technology/health-science/stress-tracking/>
 - <https://www.whoop.com/thelocker/heart-rate-variability-hrv>
 - <https://www.garmin.com/en-US/garmin-technology/health-science/sleep-tracking/>
 - <https://support.apple.com/guide/watch/track-your-sleep-apd830528336/watchos>



Stress Measurements

- Sensors measure *skin, sweat, heart beat*
- Skin (Apple, Garmin)
 - **For HRV**
 - Green light (most similar to EKG/ECG) shines on your skin and measures blood flow
 - More blood = more absorption, less reflection
 - Photodiode measures reflected light
 - Peak reflection → less blood in the capillary at that moment
 - Trough reflection → more blood during heartbeat
 - Pulse waveform is reconstructed from these peaks and troughs showing the RR intervals
 - **For blood oxygen**
 - Red and infrared light estimates blood oxygen or % of oxygen saturated hemoglobin in blood
 - The light goes into your skin (to capillaries) > some light is absorbed by blood > some reflects back
 - Oxygenated → absorbed more infrared, less red
 - Deoxygenated → absorbed more red, less infrared
 - Light reflected is measured by photodiodes
- Sources
 - https://youtu.be/_5Nh2wnChpc?si=MmsGdKONMN85K5J8
 - <https://support.apple.com/en-us/120277>



Grounding techniques

- 5,4,3,2,1
 - 5 things you see, ..., 2 things you smell, 1 thing you taste
- 1 x 30s
 - Focus on 1 stimuli for 30s
- 4x4
 - Inhale 4s, hold 4s, exhale 4s, hold 4s
- Naming (most friendly for echolalia people)
 - Name categories (favorite songs)
 - Name facts (date, time, location)
 - Recite (periodic table, times table, etc)
- “Touch grass”
 - Sitting on the ground/flat surface
 - Pace and count steps
 - Basically lower the center of gravity

Demo!

Improvements

- Slow down grounding prompts to accommodate processing speeds
- Have a settings options for the users to toggle on and off which grounding exercises they like

Name contenders

- Stressa Inte
- 걱정하지 마 geok jeong hajima
- 没事 mei shi